# Assignment 4.4

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**Task 1**

A
Classify the sentiment of the following review as Positive, Negative, or Neutral.

Review: "The delivery was late but the product works fine."

B

Classify the sentiment as Positive, Negative, or Neutral.

Example:
Review: "The product is amazing and works perfectly."
Sentiment: Positive

Now classify:
Review: "The delivery was late but the product works fine."

C

Classify the sentiment as Positive, Negative, or Neutral.

Example:
Review: "The product is amazing and works perfectly."
Sentiment: Positive

Now classify:
Review: "The delivery was late but the product works fine."

D

Classify customer reviews into Positive, Negative, or Neutral.

Review: "Great quality and fast delivery."
Sentiment: Positive

Review: "Very poor customer service."
Sentiment: Negative

Review: "Product is okay for daily use."
Sentiment: Neutral

Review: "I am extremely happy with this purchase."
Sentiment: Positive

Now classify:

Review: "The delivery was late but the product works fine."

**Output**

Neutral

**Task 2**

**1**

```
emails = [
    ("Server is down. Immediate action required!", "High
Priority"),
    ("Please review the attached report by tomorrow.", "Medium
Priority"),
    ("Reminder: Team lunch on Friday.", "Low Priority"),
    ("Client complaint about payment failure.", "High Priority"),
    ("Weekly project status update.", "Medium Priority"),
    ("Office maintenance scheduled this weekend.", "Low
Priority")
]
```

**2**

```
def zero_shot_classifier(email):
```

```
    if "urgent" in email.lower() or "down" in email.lower() or
"complaint" in email.lower():
        return "High Priority"
    elif "report" in email.lower() or "review" in email.lower() or
"update" in email.lower():
        return "Medium Priority"
    else:
        return "Low Priority"



test_email = "Server is down. Immediate action required!"
print("Zero-shot result:", zero_shot_classifier(test_email))

3

# Example provided to "AI"
example_email = "Urgent: System failure in production"
example_label = "High Priority"

def one_shot_classifier(email):
    if "urgent" in email.lower() or "failure" in email.lower():
        return "High Priority"
    elif "report" in email.lower():
        return "Medium Priority"
    return "Low Priority"



print("One-shot result:", one_shot_classifier(test_email))
```

4

```
def few_shot_classifier(email):
    keywords_high = ["urgent", "down", "failure", "complaint"]
    keywords_medium = ["report", "review", "update"]

    if any(word in email.lower() for word in keywords_high):
        return "High Priority"
    elif any(word in email.lower() for word in
keywords_medium):
        return "Medium Priority"
    else:
        return "Low Priority"


print("Few-shot result:", few_shot_classifier(test_email))
```

**Task 3**

**1**

```
queries = [
    ("How can I apply for admission?", "Admissions"),
    ("When are the semester exams scheduled?", "Exams"),
    ("I need the syllabus for Data Structures.", "Academics"),
```

```
    ("When is the campus placement drive?", "Placements"),
    ("What documents are required for enrollment?",
"Admissions"),
    ("How do I check my exam results?", "Exams")
]
```

2

```
def zero_shot_router(query):
    q = query.lower()

    if "admission" in q or "enrollment" in q:
        return "Admissions"
    elif "exam" in q or "result" in q:
        return "Exams"
    elif "syllabus" in q or "subject" in q:
        return "Academics"
    elif "placement" in q or "job" in q:
        return "Placements"
    else:
        return "Unknown"


test_query = "When is the campus placement drive?"
print("Zero-shot:", zero_shot_router(test_query))
```

3

```
# Example given to AI
example_query = "Admission form submission deadline?"
example_label = "Admissions"

def one_shot_router(query):
    q = query.lower()

    if "placement" in q:
        return "Placements"
    elif "exam" in q:
        return "Exams"
    elif "syllabus" in q:
        return "Academics"
    elif "admission" in q:
        return "Admissions"
    return "Unknown"



print("One-shot:", one_shot_router(test_query))
```

4

```
def few_shot_router(query):
    q = query.lower()

    admissions_keywords = ["admission", "enrollment", "apply"]
    exam_keywords = ["exam", "result", "hall ticket"]
```

```
    academic_keywords = ["syllabus", "course", "subject"]
    placement_keywords = ["placement", "internship", "job"]

    if any(word in q for word in admissions_keywords):
        return "Admissions"
    elif any(word in q for word in exam_keywords):
        return "Exams"
    elif any(word in q for word in academic_keywords):
        return "Academics"
    elif any(word in q for word in placement_keywords):
        return "Placements"
    else:
        return "Unknown"


print("Few-shot:", few_shot_router(test_query))
```

Output

```
Zero-shot: Placements
One-shot: Placements
Few-shot: Placements
```